Optical Knowledge Assisted Unsupervised Cross-domain SAR Target Detection

***Yu Shi*[1], *Lan Du*[1*], *Yuchen Guo*[2], *Yuang Du*[1]**

[1]*National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an, China*
[2]*Academy of Advanced Interdisciplinary Research, Xidian University, Xi'an, China*
**dulan@mail.xidian.edu.cn*

Keywords: SAR Target detection, Cross-domain, Optical knowledge, Generative adversarial networks (GANs), pseudo-labels.

Abstract

Convolutional neural networks (CNNs) based synthetic aperture radar (SAR) target detection methods have garnered attention in recent years. However, labeled SAR images are scarce and expensive to acquire, making training CNNs challenging. In this paper, an unsupervised cross-domain SAR target detection method with the assistance of the knowledge of the labeled optical domain is proposed. Our method gradually improves the performance of the detector in the unlabeled SAR target domain from three stages. At the first stage, to reduce visual dissimilarity between optical and SAR images, the Optical-to-SAR generative adversarial network (GAN), namely Opt2SAR GAN with skip layer is proposed to generate labeled fake SAR images. At the second stage, the adversarial learning strategy is employed to make the detector learn the domain invariant features between the two domains. At the last stage, to alleviate domain-bias problem, we use pseudo-labels predicted by the detector of the second stage to learn more discriminative representations of the SAR domain. To further alleviate the impact of wrong pseudo-labels, we select confident pseudo-labels according the image uncertainty and proposal uncertainty. Experiments on two cross-domain detection datasets of vehicles and ships demonstrate the effectiveness of our method.

1 Introduction

Synthetic aperture radar (SAR) can provide high-resolution images with the condition of all-day, all-weather, and long-range, and has played an important role in military and civilian fields since its birth. As an important branch of SAR image interpretation, SAR target detection has garnered attention in recent years [1].

With the rapid development of deep learning technologies, convolutional neural network (CNN) based SAR target detection methods have achieved remarkable performance with a large amount of labeled training data. However, for SAR images, it is costly to annotate objects since they are difficult to be understood intuitively, making training deep detectors with labeled SAR images challenging [2]. As an alternative way, unsupervised domain adaptation can achieve unsupervised cross-domain target detection by transferring from an easy-to-label source domain to a hard-to-label target domain. In this paper, we aim to perform unsupervised cross-domain SAR target detection with the assistance of the knowledge of the labeled optical domain. In this way, we can achieve good detection performance without incurring huge labeling costs.

In the computer vision fields, unsupervised cross-domain target detection has been extensively and intensively studied. One such method was introduced by Chen et al. [3], who utilized adversarial learning to align the distribution of features between two domains at both the image and proposal feature levels. Building upon this work, multi-alignment was proposed by Shen et al. [4], which aimed to obtain a cross-domain detector at multi-level feature space. Z. He [5] shared a similar idea. Chen et al. [6] weigh the source domain features by measuring the similarity between source domain features and target domain features. Kim et al. [5] proposed a new paradigm with multiple interpolation domains. In terms of SAR target detection, Shi et al. [7] proposed for the first time a domain-adaptive detection method across radar sensors.

The proposed domain-adaptive object detection methods point us to a feasible path. We aim to achieve unsupervised SAR target detection with the aid of annotated optical data domains. Compared with SAR images, optical remote sensing images are easier to understand intuitively, and it is easy to rely on the data labeling platform to obtain large-scale annotated data to train the detection network. In this paper, an unsupervised cross-domain SAR target detection with the assistance of the knowledge of the labeled optical domain is proposed. Our method gradually improves the performance of the detector in the unlabeled SAR target domain from three stages. At the first stage, to reduce the visual dissimilarity between labeled optical and unlabeled SAR images, the designed generative adversarial network (GAN), namely Opt2SAR GAN with skip layer is proposed. At the second stage, the adversarial learning strategy is employed to make the detector learn the domain invariant features between the two domains. At the last stage, to alleviate the domain-bias problem, we use the pseudo-labels predicted by the detector of the second stage to learn more discriminative representations of the SAR domain. To further alleviate the impact of wrong pseudo-labels, we select confident pseudo-labels according to the image uncertainty and proposal uncertainty. More specifically, the main contributions can be concluded as follows.

1) To alleviate the visual dissimilarity between optical and

https://doi.org/10.1049/icp.2024.1303

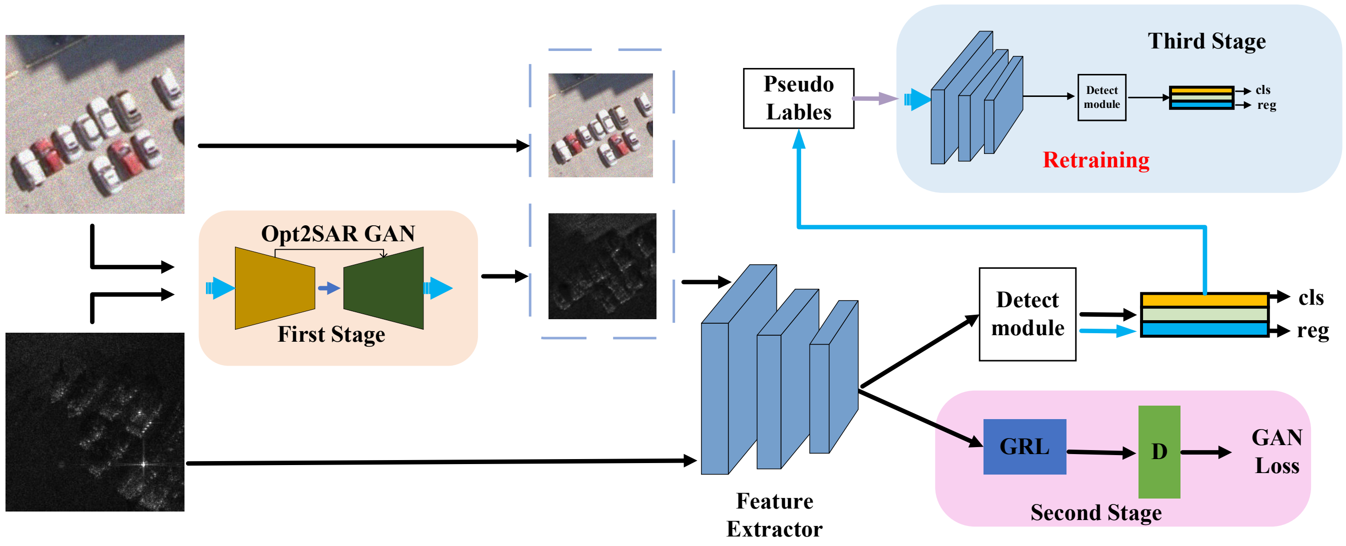


Fig. 1. The overall framework of the proposed unsupervised cross-domain SAR target detection with the assistance of the knowledge of the labeled optical domain. Our method gradually improves the performance of the detector in the unlabeled SAR target domain from three stages, including Opt2SAR GAN, adversarial alignment with GRL, and pseudo-label retraining.

SAR images, the Optical-to-SAR generative adversarial network (GAN), namely Opt2SAR GAN with skip layer is proposed.

2) We propose a discriminative learning approach for the target SAR domain that combines image-level and proposal-level pseudo-label filtering.

3) Experiments on two optical-SAR cross-domain detection datasets of vehicles and ships demonstrate the effectiveness of our method.

Next, our proposed unsupervised cross-domain detection will be given in Section II. The Experiments results and analysis on two optical-SAR cross-domain detection datasets of vehicles and ships will be given. Finally, Section IV gives the conclusion.

2 Methodology

The overall framework of the proposed unsupervised cross-domain SAR target detection with the assistance of the knowledge of the labeled optical domain is shown in Fig. 1. Our method gradually improves the performance of the detector in the unlabeled SAR target domain from three stages. In the first stage, the unpaired optical and SAR images are used to train the Opt2SAR GAN and we then translate the labeled optical images to fake SAR images. We can get a labeled intermediate domain with little difference from the SAR domain. In the second stage, we further apply adversarial alignment with gradient reverse layer (GRL) [9] to reduce the feature difference between the intermediate and SAR domains. In the last stage, we use the pseudo-labels predicted by the detector of the second stage to learn more discriminative representations of the SAR domain. To further alleviate the impact of wrong pseudo-labels, we select confident pseudo-labels according to the image uncertainty and proposal uncertainty.

2.1 First Stage: Opt2SAR GAN

It is well known that there are huge differences in the appearance of optical and SAR data since they have differences in imaging mechanisms. To reduce the appearance

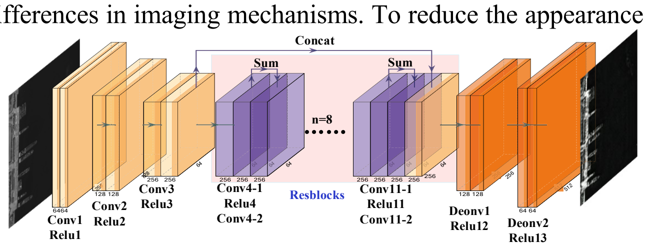


Fig. 2. The network of our generator, which consists of the encoder, decoder, and skip-connection.

difference, we aim to design a GAN to translate the labeled optical images into fake SAR images. We can get a labeled intermediate domain with little difference from the SAR domain. Then, the detection model trained on the labeled intermediate domain can perform well in the SAR domain.

Opt2SAR GAN consists of a generator G and a discriminator D, both of which are trained in an adversarial learning manner. The generator aims to create fake SAR images in an attempt to deceive the discriminator, whereas the discriminator aims to differentiate between genuine SAR data and the fake SAR data generated by the generator (in the intermediate domain). The adversarial loss is as follows:

$$\begin{aligned}\mathcal{L}_{GAN}(G_{o\to S}, D_S, X_o, X_S) &= \mathrm{E}_{x_S\sim X_S}[\log D_S(x_S)] \\ &+ \mathrm{E}_{x_o\sim X_o}[\log(1-D_S(G_{o\to S}(x_o)))]\end{aligned} \tag{1}$$

Where $G_{o\to S}$ represents the generator that generats from optical images to SAR images and D_S represents the discriminator that identifies the SAR images.

Due to the significant differences between the two domains, generating high-quality SAR images presents a challenge for the generator. While optical images contain a range of visual features, SAR images are primarily characterized by their texture, structure, and shape, which are mostly captured in the low layers of convolutional neural networks. To tackle this problem and ensure that the generator produces realistic SAR images, we incorporated a skip connection into the generator's network. The architecture is shown in Fig. 2. Drawing from previous work, we made improvements to the generator's architecture, which includes three down-sampling layers (each layer consists of a ‘conv’ operation and ‘Relu’ operation), eight residual blocks (each block consist of 2 ‘conv’ operations and a ‘Relu’ operation, using the skip-line to connect blocks), and two up-sampling layers (each layer consists of a ‘deconv’ operation and ‘Relu’ operation). Specifically, we incorporated a concat-operation skip connection between the third and third-to-last layers to preserve the low-level texture, structure, and shape information.

While the GAN loss enables the generator to generate data that looks realistic, it doesn't ensure that the generator preserves the arrangement of images or the position and orientation of targets. To address this challenge, we employ a cycle-consistency constraint, which is commonly used in image-to-image translation methods [10]. This constraint involves training another generator and another discriminator, using the same GAN loss as the original generator and discriminator. The cycle-consistency constraint forces the generators to produce SAR images that can be translated back to the original optical images. It ensures that the generators satisfy both Optical-to-SAR and SAR-to-Optical translation requirements. We define the cycle-consistency loss as:

$$\begin{aligned}\mathcal{L}_{cycle}(G_{o\to S}, G_{o\to S}, X_o, X_S) = \\ \mathrm{E}_{x_o\sim X_o}[\| G_{S\to o}(G_{o\to S}(x_o) - x_o \|_1] \\ + \mathrm{E}_{x_S\sim X_S}[\| G_{o\to S}(G_{S\to o}(x_S) - x_S \|_1]\end{aligned} \tag{2}$$

The final loss is as follows:

$$\begin{aligned}\mathcal{L} = \lambda \mathcal{L}_{GAN}(G_{o\to S}, D_S, X_o, X_S) \\ + \mathcal{L}_{GAN}(G_{S\to o}, D_o, X_o, X_S) \\ + \eta \mathcal{L}_{cycle}(G_{o\to S}, G_{o\to S}, X_o, X_S)\end{aligned} \tag{3}$$

where λ and η are balance parameters.

2.2 Second stage: feature-level alignment with adversarial learning

To bridge the feature gap between the two domains, we aim to reduce the feature difference. Since the labeled intermediate domain is the only source of labels, a feasible idea is to extract the common features of the two domains, then the labeled intermediate domain can still guide the detector to detect SAR images effectively. To achieve this goal, feature-level alignment with adversarial learning is applied to extract domain-common features. Given the significant differences between the two domains, we align distributions only at the high semantic space that can reflect the object's semantic information, such as the last convolutional layers in feature extractors. To this end, we use domain adversarial learning. Specifically, in the domain adversarial learning framework, given the high semantic feature, as well as a domain classifier D, D aims to classify whether the given feature belongs to the SAR or the intermediate domain, while simultaneously training the detector to deceive the domain classifier. The above processes are described as follows:

$$\begin{aligned}\mathcal{L}_{adv}(F, D, X_{inter}, X_S) = \mathrm{E}_{x_{inter}\sim X_{inter}}[\log D(F(x_{inter}))] \\ + \mathrm{E}_{x_S\sim X_S}[\log(1 - D(F(x_S)))]\end{aligned} \tag{4}$$

Where X_{inter} represents the intermediate image and X_S represents the SAR image. F represents the feature extractor and is used to obtain the high semantic features.

In this framework, the domain classifier is trained to maximize an objective, while the feature extractor is just the opposite. To connect the two, we use the gradient reversal layer (GRL), which flips the sign of the gradient during backpropagation. By using the GRL, the feature extractor is forced to learn domain-common semantic features, while the domain classifier is trained to accurately classify the features based on their domain of origin. The feature extractor tends to extract features that are common to both domains with this adversarial training and thus can be used to guide the detector to detect SAR images effectively.

The detection loss is the same as Faster RCNN :

$$\begin{aligned}\mathcal{L}_{det} = \mathcal{L}_{cls} + \mathcal{L}_{loc} \\ = \mathcal{L}_{cls}^{rpn} + \mathcal{L}_{loc}^{rpn} + \mathcal{L}_{cls}^{roi} + \mathcal{L}_{loc}^{roi}\end{aligned} \tag{5}$$

The overall objective loss of the feature level can be written as:

$$\mathcal{L} = \mathcal{L}_{det} + \alpha \mathcal{L}_{adv} \tag{6}$$

where α is the balance parameter.

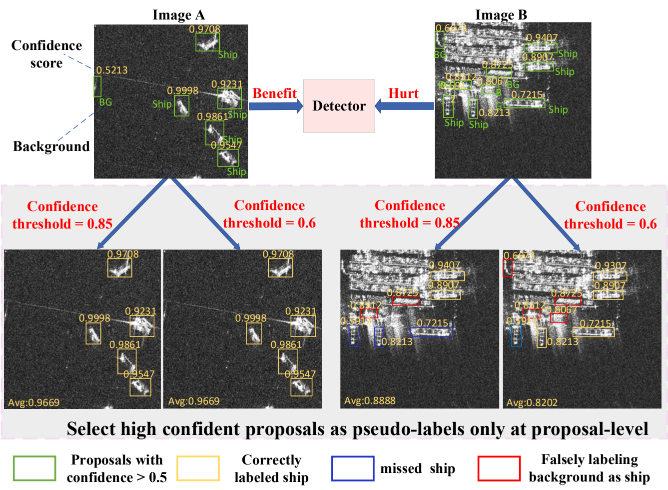
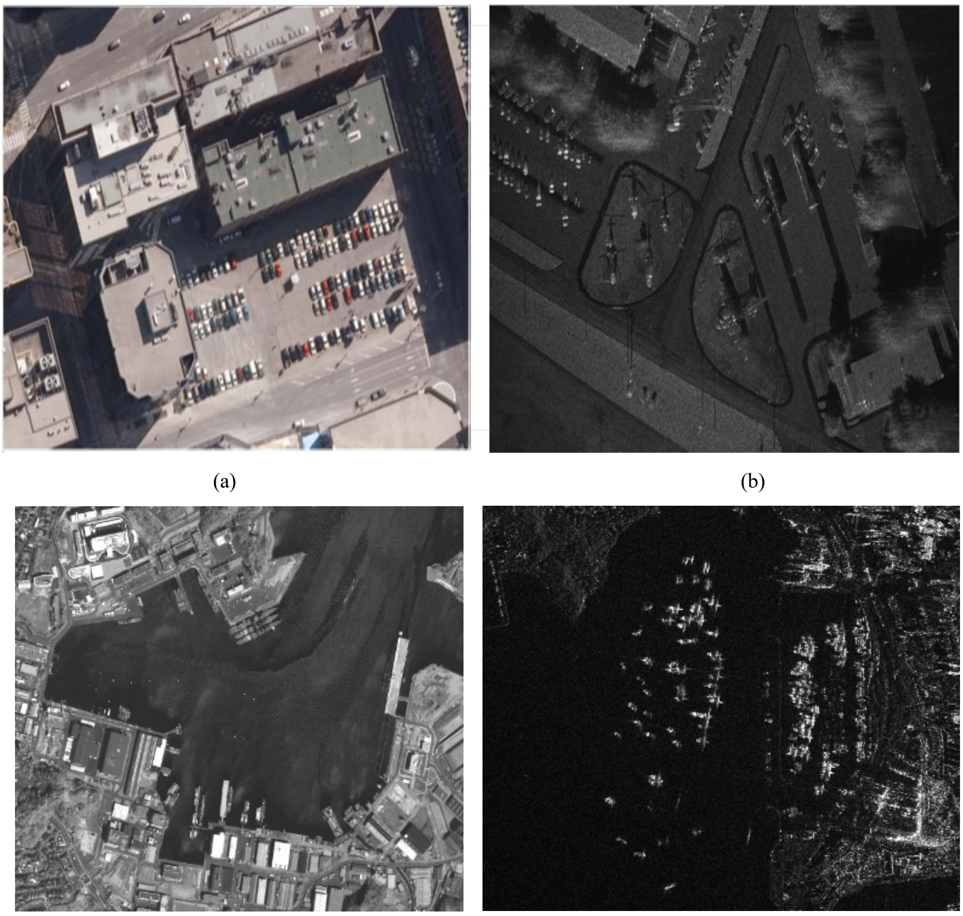


Fig. 3. In Image A, there are more correct pseudo-labels, whether a high or low confidence threshold is used. However, in Image B, there are more incorrect pseudo-labels. Setting a high threshold may result in high missed detections, while setting a low threshold may cause high false alarms. As a result, Image B should not be selected for training the detector.

2.3 Third stage: pseudo-labels learning

In the previous stage, the domain-common feature representation was used to align high-level feature distributions, but it may overlook domain-specific features that could aid target detection. To address this, we use the pseudo-labels predicted by the detector of the second stage to learn more discriminative representations of the SAR domain. To further alleviate the impact of wrong pseudo-labels, we select confident pseudo-labels according to the image uncertainty and proposal uncertainty. In addition, we select high-quality pseudo-labels at both proposal and image levels and introduce label smoothing to enhance fault tolerance.



(a) (b) (c) (d)

Fig. 4. Examples of the optical and SAR remote sensing images. (a) and (b) are the car data. (a) the example of the Toronto car dataset, (b): the example of the MiniSAR car dataset. (c) and (d) are the ship data. (c) the example of the GF2 ship dataset, (d): the example of the AIR-SARship-1.0 ship dataset. It can be seen from the figure that optical images have richer semantic information.

Pseudo-labeling techniques are widely used in semi-supervised learning tasks[12][13], and picking high-quality pseudo-labels is crucial for detector learning. Most previous methods only consider high-confidence proposals, but ignore the overall quality of pseudo-labels in an image. As shown in Fig. 3, for proposal selection, setting a high threshold on difficult images may lead to treating targets as backgrounds, while setting a low threshold may lead to treating backgrounds as targets. In contrast, our proposed method takes into account the quality of the pseudo-labels at both the proposal and image levels. We classify images as easy or difficult based on the number of correct or wrong pseudo-labels they have, respectively. Setting a high threshold on difficult images may lead to treating ships as backgrounds, while setting a low threshold may lead to treating backgrounds as targets. To mitigate this problem, we select easy images at both levels by first setting a confidence threshold on proposals and then calculating the average confidence score of all selected proposals inside an image.

Proposal-Level selection:

$$\hat{y}_j = \begin{cases} 1, & p_j \geq thresh. \\ 0, & Otherwise. \end{cases} \tag{7}$$

We only select the proposals whose confidence probability exceeds a threshold.

Image-Level selection:

$$score_i = \frac{1}{m_i} \sum_{j=1}^{m_i} p_{i,j}. \tag{8}$$

For one SAR image with m proposals, we calculate the image score according to the mean confidence probability of all proposals.

We learn iteratively: fix detector parameters, predict pseudo-labels; fix labels, and update detector parameters. We calculate the average confidence score of all images from the SAR domain and then sort them according to their average score from high to low. In the first iteration round, we select the top 50% of SAR images with image scores to retrain the detector, and then further use the modified detector to predict pseudo-labels. In the second and third iteration rounds, we select extra 20% SAR images each time.

3 Experiments

In this section, experiments on two optical-SAR cross-domain detection datasets of vehicles and ships will be given.

TABLE I. COMPARISON RESULTS ON CAR DATASET

	Precision	Recall	F1-score
Gaussian CFAR	0.3587	0.8048	0.4962
DAF	0.7333	0.1789	0.2876
HTCN	0.4494	0.5691	0.5022
Proposed method	0.7760	0.7805	**0.7782**
Faster RCNN-supervised	0.8421	0.8943	0.8674

TABLE II. COMPARISON RESULTS ON SHIP DATASET

	Precision	Recall	F1-score
Gaussian CFAR	0.4014	0.9040	0.5559
DAF	0.4792	0.1840	0.2659
HTCN	0.6061	0.4800	0.5357
Proposed method	0.7669	0.8160	**0.7907**
Faster RCNN-supervised	0.7483	0.8800	0.8122

3.1 Dataset

Optical2SAR ship dataset: The optical ship data are collected by GF2 satellite. The optical images include 5high-resolution images with dimensions of 29200×27620 and 37 lower-resolution images with dimensions of 4000×4000, all with a resolution of 0.8 m. To create a source domain for our study, we cropped the optical images into 1280 sub-images of size 512×512. The AIR-SARship-1.0 dataset, on the other hand, consists solely of SAR images, with 31 images in total and 879 ships. The images have resolutions of 1 m or 3 m. The training set is made up of 374 sub-images cropped from 21 original SAR images of size 3000×3000, while the test set contains the remaining 10 original images of the same size. Fig. 4 shows an example of a GF2 image and an AIR-SARship-1.0 image. The scenes and target information in optical images are less expensive to be interpreted by person, which means it is easy to rely on the data labeling platform to obtain large-scale annotated data to train the detection network.

1) *Optical2SAR car dataset:* The Toronto measured data set is a kind of measured data of Toronto urban remote sensing optical images based on complex scenes, including 11 RGB images with a size of pixels 500×7500 and a resolution of 0.15m×0.15m.The MiniSAR measured data set is the SAR image measured data based on the complex scene released by the Sandia National Laboratory of the United States in 2006. In the experiment, 9 SAR images were used, the image size is 1638×2510, and the resolution is 0.1m×0.1m. 7 MINISAR images are used to train the model and 2 are used to test the model. Toronto and MiniSAR images contain complex backgrounds, such as buildings, grass, trees, etc., and only vehicles are the objects of interest. Each image usually contains multiple vehicle targets, and the size of the vehicle targets is not very different, but the size of the vehicle targets between different datasets has a large gap. Fig. 4 shows an example of a Toronto image and a MiniSAR image.

3.2 Evaluation index

The specific calculation formulas for performance indicators are shown in (9) and (10). In these formulas, TP represents true positives, which are the number of detection boxes where the detection value is true and the annotation value is also true. FP represents false positives, which are the number of detection boxes where the detection value is true and the annotation value is false. NP represents false negatives, which are the number of detection boxes where the detection value is false and the annotation value is true.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{Recall} = \frac{\text{TP}}{\text{NP}} \tag{10}$$

3.3 Performance Comparison

We conducted experiments to compare our approach with three other unsupervised detection methods, including the conventional CFAR method, namely Gaussian-CFAR [11], and two other UDA methods [3] and [6] for target detection, namely DAF and HTCN. We also compared our approach with a fully-supervised Faster R-CNN method.

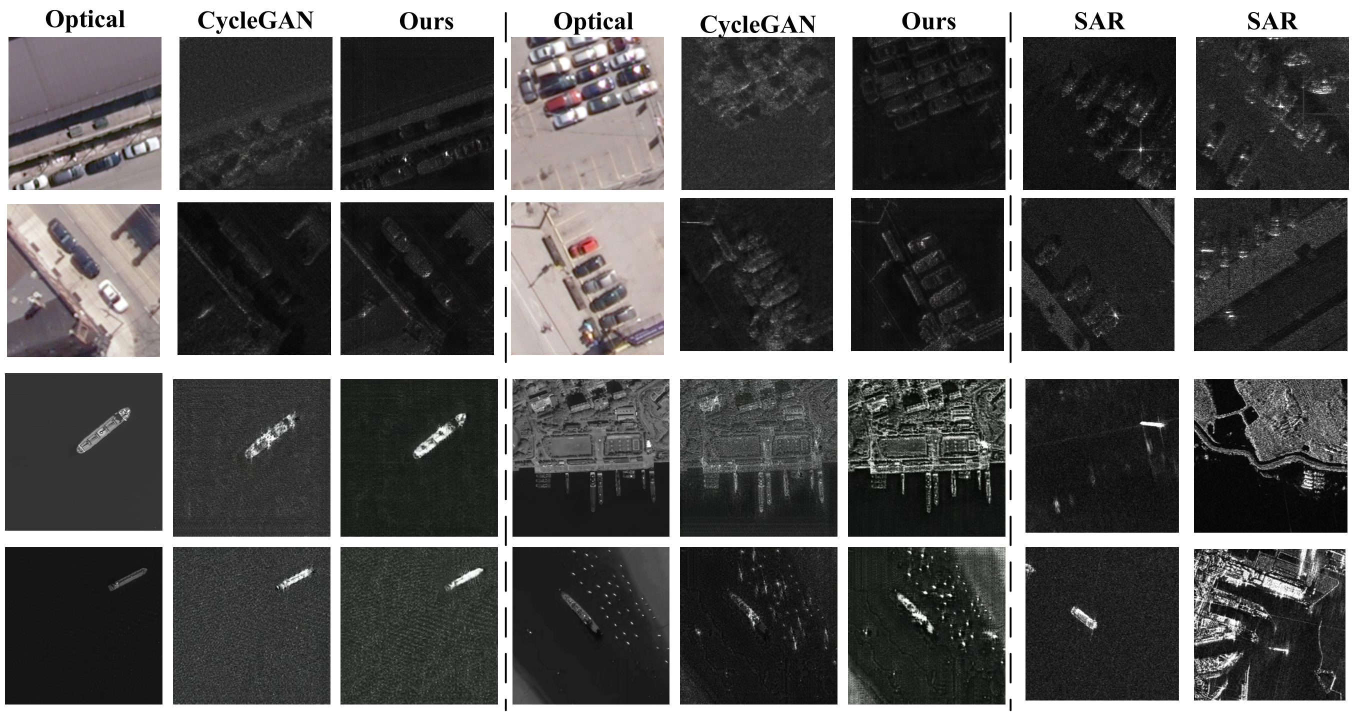


Fig.5. The results of Opt2SAR GAN and compare them with those of CycleGAN. The original images and the transition results of two GANs are given on the left and the middle, and the real SAR image is on the right.

Table I and Table II give the domain adaptation results of the car dataset and ship dataset. As shown in Table I and Table II, the results indicate that Gaussian-CFAR performs poorly compared to the other methods, especially in complex scenes. The performance of the two unsupervised domain adaptation detection methods, DAF and HTCN, was even worse than Gaussian-CFAR. Among the many methods, the detection performance of DAF is the worst because it tries to reduce the difference between optical and SAR only at the feature space. HTCN uses CycleGAN to reduce pixel-level differences but limits the improvement of detection performance by aligning feature distributions at multiple features. Our proposed method achieved the best detection performance among all methods. The experimental results of the vehicle dataset and the ship dataset show that the vehicle unsupervised domain adaptation detection task is more difficult than the ship detection task. The performance gap between the proposed method and the fully supervised method on the vehicle dataset is significantly larger than that on the ship dataset. This is because the background and clutter of the vehicle are more complex.

3.4 Visualization

We show the visualized results of Opt2SAR GAN in Fig. 5 and compare the generation results of our method with those of CycleGAN. In Fig.5, the original images and the transition results of two GANs are given on the left and the middle, and the real SAR image is on the right. Although optical and SAR images observe similar scenes, they differ greatly in style and describe objects in different ways. The GAN method reduces the gap between them by cross-domain transformation. From Fig. 5, we observe that the images generated by Opt2SAR GAN are highly similar to actual SAR images. Moreover, our OSGAN-generated images better preserve the edge, structure, and shape information when compared to CycleGAN, which highlights the effectiveness of our proposed approach. Since the sea scene is simpler than the land scene, the generation effect of the ship target is more realistic than that of the vehicle.

4 Conclusion

This article explores the feasibility of using labeled optical cognition knowledge to assist the task of unsupervised SAR detection and proposes a feasible idea for the development of the community. Using GAN to convert optical images of similar scenes into SAR images shows strong feature correlation and inter-domain transferability between the two domains. The learning of inter-domain correlation features further narrows the difference between the two data domains. The training with pseudo labels also is efficient, especially by selecting confident pseudo labels according to the image uncertainty and proposal uncertainty.

The resolution of the optical data and SAR data in our work is not much different, and future work should continue to explore the transfer learning problem between data domains with different resolutions. Using optical knowledge to achieve cross-domain unsupervised SAR target recognition is also a point that needs to be studied in the future.

5 Acknowledgements

This work was supported in part by the National Science Foundation of China under Grant U21B2039, in part by the 111 Project.

6 References

[1] B. Hou, X. Chen and L. Jiao, "Multilayer CFAR detection of ship targets in very high resolution SAR images", IEEE Geosci. Remote Sens. Lett., vol. 12, no. 4, pp. 811-815, Apr. 2015.

[2] J. He et al., "A Cross-Modality Feature Transfer Method for Target Detection in SAR Images," in IEEE Trans. Geosci. Remote Sens., vol. 61, pp. 1-15, 2023.
[3] Y. Chen, et al, "Domain adaptive faster R-CNN for object detection in the wild," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3339–3348.
[4] Z. Shen, et al, "SCL: Towards Accurate Domain Adaptive Object Detection via Gradient Detach Based Stacked Complementary Losses," arXiv: 1911.02559, 2019,
[5] Z. He et al, "Multi-adversarial faster-RCNN for unrestricted object detection," in Proc. IEEE Int. Conf. Comput. Vision, 2019, pp. 6668–6677.
[6] C. Chen, et al, "Harmonizing transferability and discriminability for adapting object detectors," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8869–8878.
[7] Y. Shi, et al, "Unsupervised Domain Adaptation for SAR Target Detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6372-6385, June 2020.
[8] T. Kim, M. Jeong, et al, "Diversify and match: A domain adaptive representation learning paradigm for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 12456–12465.
[9] Y. Gain, and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in Proc. Int. Conf. Mach. Learn., 2015, pp. 1180–1189.
[10] J. Zhu, T. P. I. Park, and A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in Proc. IEEE Int. Conf. Comput. Vis., 2017, pp. 2223–2232.
[11] G. Gao, L. Liu, L. Zhao et al., "An adaptive and fast CFAR algorithm based on automatic censoring for target detection in high-resolution SAR images," IEEE Trans. Geosci. Remote Sens., vol. 47, no. 6, pp. 1685–1697, Jun. 2009.
[12] D.-H. Lee, "Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks," in Proc. Workshop Challenges Represent. Learn., 2013, vol. 3, no. 2, pp. 1–6.
[13] K. Sohn et al., "Fixmatch: Simplifying semi-supervised learning with consistency and confidence," in Proc. Adv. Neural Inf. Process. Syst., 2020, pp. 596–608.